

Draft AV: Behavioral Surplus as Training Data — How Social Platforms Extract, Repurpose, and Monetize User-Generated Content for AI

Version: v0.4 | **Status:** Draft — all platform-statistics & licensing-pricing markers resolved (1 → 0 remaining); Bommasani 2024 tag corrected to [P] **Handbook:** Springer Handbook of Hybrid Intelligence **Related drafts:** F (commons extraction), C (training data memorization), AF (human labor in AI training), W (unharvested commons), N (AI/ML ecosystem), Q (machines using humans)

Abstract

The largest AI training datasets in existence are not academic corpora or curated open-source releases — they are the accumulated behavioral and content records of billions of users of commercial social platforms. Meta, Google/YouTube, X (formerly Twitter), LinkedIn, TikTok, and Reddit collectively hold text, image, video, audio, behavioral interaction, and social graph data at a scale that dwarfs every open research dataset combined. This chapter examines how these platforms have transitioned from using user data for recommendation and advertising targeting to using it as training substrate for generative AI foundation models — and how this transition represents a qualitatively distinct extraction that existing privacy, copyright, and competition frameworks were not designed to govern. Drawing on Zuboff's surveillance capitalism framework, the legal analysis of ToS-as-consent under GDPR, enforcement actions by the Irish Data Protection Commission and UK ICO, and the emerging regulatory gap in the EU AI Act's treatment of GPAI systems, the chapter argues that social platform data constitutes a second layer of extraction beyond the open-knowledge-commons extraction documented in Draft F: not merely the collective product of volunteer contributors, but the intimate behavioral record of billions of users who were never asked whether their interactions could train AI systems that would subsequently be commercialized or deployed against their interests.

§1 The Scale of the Asset

The quantitative gap between open training datasets and social platform data is not marginal — it is structural. The Pile (EleutherAI, 2020), among the largest openly documented pre-training corpora, contains approximately 825 GB of deduplicated text. Common Crawl, the nonprofit web snapshot that underlies most major LLMs, holds approximately 9.5 petabytes. Against these, consider:

- **Meta** holds the social graph, posts, comments, reactions, messages, photos, and videos of approximately 3.43 billion Family daily active people across Facebook, Instagram, WhatsApp, and Threads (Meta, 2025a — Q1 2025 earnings, March 2025 average). The Instagram photo corpus alone exceeds tens of billions of images. WhatsApp processes more than 100 billion messages per day (Meta Platforms, Inc., 2020 — Q3 2020 earnings call, 29 October 2020, Zuckerberg disclosure).
- **Google/YouTube** hosts a public-video corpus exceeding several hundred million videos, with combined watch time surpassing one billion hours per day (Wojcicki, 2017 — YouTube Official Blog) and new content uploaded at approximately 500 hours per minute as of 2019 (Hale, 2019 — Tubefilter, citing YouTube data). Google Search has processed over 2 trillion queries per year since at least 2016 (Sullivan, 2016 — Search Engine Land, citing Google disclosure), with each query linked to a behavioral trail.
- **X (Twitter)** generated approximately 500 million posts per day at its peak under the Twitter brand (Krikorian, 2013 — Twitter Engineering Blog). The platform's value proposition to AI developers rests largely on its real-time text corpus — the only near-complete record of public human discourse over 15 years.

- **LinkedIn** holds professional profiles, career histories, job applications, professional content, and messaging data for more than 1 billion members (Roslansky, 2023 — LinkedIn News, 2 November 2023; Microsoft FY24 Q1 earnings, October 2023).
- **TikTok** had more than 1 billion monthly active users as of September 2021 (TikTok, 2021 — TikTok Newsroom, "Thanks a billion!"), with subsequent third-party analyst estimates placing the figure above 1.5 billion by 2024 (Statista, 2024 — analyst aggregation, no official disclosure since 2021), and holds video, audio, interaction, and behavioral data in quantities that rival YouTube while being substantially less scrutinized by Western regulators due to its Chinese ownership.
- **Reddit** hosts more than 1 billion posts and 16 billion comments accumulated since 2005, with averages of 1.2 million posts and 7.5 million comments daily as of 2023 (Reddit, Inc., 2024 — S-1 Registration Statement) — a text corpus notable for its depth of technical, scientific, and vernacular knowledge, and one of the few platform datasets for which a public licensing market has emerged.

This data is categorically different from the open-knowledge commons. Wikipedia and arXiv contain knowledge that contributors deliberately made public for knowledge-sharing purposes. Social platform data contains behavioral traces, private communications, medical disclosures, political views, relationship history, and professional information that users shared within an assumed context — with friends, professional networks, or public followers — not with AI training pipelines. The legal concept of **contextual integrity** (Nissenbaum, 2004) holds that information flows appropriately when they match the norms of the context in which information was originally shared; training a commercial AI foundation model on intimate personal disclosures violates contextual integrity even when the data was technically "public."

§2 From Recommendation to Generation: A Qualitative Shift

These platforms have always trained machine learning models on user data. The recommender systems that determine which content each user sees, which advertisements are served, and which connections are suggested have been trained on behavioral data since at least 2010 (Facebook's early EdgeRank algorithm; YouTube's 2016 deep neural network recommender). This use was, in most jurisdictions, largely uncontested: it was understood as core to the platform's value proposition and explicitly anticipated in terms of service.

The transition to **generative AI foundation model training** is qualitatively different in three respects.

Scale of inference. Recommendation models predict preferences for content the user already produces; they operate within a closed personalization loop. Foundation models trained on the same data can generate arbitrary text, images, and behavior in any context, including contexts the original data producer never anticipated. A model trained on Reddit discussions can produce legal advice, medical guidance, or political rhetoric — capabilities that were not implicit in the original act of contributing to an online forum.

Externalization of the model. Platform recommendation systems remain within the platform's own infrastructure, shaping only what the contributing user sees. A foundation model trained on platform data and deployed externally — as a commercial API, a standalone product, or a third-party integration — takes the extracted value entirely outside the platform relationship. The contributing user receives no ongoing benefit; the trained model can operate indefinitely without further data contribution.

Competitive displacement. Models trained on social platform data can replicate the epistemic functions of the platforms themselves. A language model trained on Stack Overflow's corpus can answer programming questions without directing traffic to Stack Overflow; one trained on Reddit can provide community discussion without Reddit's ad revenue. The extractive relationship documented in Draft F (training on GitHub to build Copilot; training on Stack Overflow to reduce Stack Overflow traffic) applies with even greater force to social platform corpora, because social platform value is constituted precisely by text and behavioral data.

§3 Platform Practices: What Is Known

3.1 Meta

Meta's use of user data for AI training became a matter of public controversy and regulatory action in 2023–2024. The chronology is instructive.

February 2023: Meta released the first Llama model as a research artifact (Touvron et al., 2023). Subsequent Llama generations (Llama 2 in July 2023, Llama 3 in 2024) progressively expanded both model capability and the disclosed training-data envelope, which Meta has consistently described as "publicly available data from the internet."

May–June 2024: Meta began notifying EU and UK users that, effective 26 June 2024, public posts, comments, captions, and Stories from adult Facebook and Instagram users would be used to train Meta's generative AI systems under a "legitimate interests" basis (GDPR Article 6(1)(f)). The advocacy group noyb filed eleven complaints across EU jurisdictions, and the UK Information Commissioner's Office and Irish Data Protection Commission engaged Meta directly.

14 June 2024: The Irish Data Protection Commission (DPC), Meta's lead supervisory authority in the EU under GDPR's one-stop-shop mechanism, announced that Meta had agreed to **pause AI training on EU and UK user data** following intensive engagement (Data Protection Commission, 2024). Meta complied, creating a documented two-tier data use policy: EU/UK users' data excluded; non-EU users' data included. This was the most direct regulatory intervention to date in foundation model training on social platform data.

April 2025: Meta announced it would resume EU AI training following further engagement with the DPC and an EDPB Article 64 Opinion (December 2024) on the conditions under which legitimate interests could ground GenAI training. Multiple EU data protection authorities and advocacy groups (including noyb) continued to challenge the basis, arguing that users should have the right to opt in before training commences rather than opt out after.

2024–2025: Meta's AI Assistant, powered by Llama models, was integrated directly into WhatsApp, Instagram, Messenger, and Facebook — completing the loop from data extraction to deployment within the same platform ecosystem. The competitive displacement concern is acute: a model trained on WhatsApp conversations deployed as WhatsApp AI creates a recursive dependency in which users' past conversations train the system they interact with today.

Meta's Llama model family explicitly acknowledges in technical documentation that it was trained in part on "publicly available data from the internet" — a phrase that, in Meta's context, includes public Facebook and Instagram posts. Full training data composition has not been publicly disclosed in the manner of Brown et al. (2020) for GPT-3.

3.2 X (Twitter / xAI)

Twitter's role as an AI training corpus has a longer history than is commonly recognized. Under its previous ownership, Twitter's "Firehose" API — providing real-time access to all public tweets — was licensed to research institutions and commercial data resellers under agreements that explicitly prohibited use for AI training without additional authorization.

October 2022: Elon Musk's acquisition of Twitter (rebranded X in 2023) preceded a rapid transition in data access policy. Within months, the academic research API was closed, the free API was restricted, and API access pricing was raised to levels that effectively excluded academic and nonprofit use.

July 2023: xAI, Musk's new AI company, was founded. Grok, xAI's large language model, was announced as having access to real-time X data as a competitive differentiator — the only LLM with live access to the full

public conversation stream.

2023–2024: X's data licensing program was restructured into a tiered Enterprise API (Perez, 2023 — *TechCrunch*, 29 March 2023; Coalition for Independent Technology Research, 2023, 3 April), with enterprise access ranging from \$42,000 to \$210,000 per month — i.e., approximately \$0.5M to \$2.5M per year for bulk firehose access — and AI training data emerging as a product category alongside traditional firehose use cases. The effective result was that a corpus previously available for academic research was privatized, consolidated under a single AI company (xAI, founded July 2023; X formally acquired by xAI in March 2025), and commercialized.

The legal basis for Grok's training on X user data has not been publicly analyzed by a regulatory authority. The US regulatory environment, unlike the EU, does not impose equivalent pre-training transparency requirements.

3.3 LinkedIn

LinkedIn's September 2024 handling of AI training consent became a textbook case in regulatory pushback speed.

18 September 2024: LinkedIn updated its privacy settings to include a new opt-out toggle for "Data for Generative AI Improvement" — a setting **enabled by default**, meaning all LinkedIn users' content and behavioral data was opted into AI training unless they actively navigated to settings and unchecked it. The opt-out covered LinkedIn's use of profile data, posts, and engagement signals to train generative AI models, including but not limited to Microsoft's broader AI infrastructure.

The rollout was disclosed only in an updated privacy notice and settings page, not through proactive notification. Technology journalists and privacy advocates noticed within hours; the backlash was immediate.

20 September 2024: the UK Information Commissioner's Office (ICO) issued a public statement that LinkedIn had "suspended such model training pending further engagement with the ICO" — a turnaround of approximately two days from the original rollout to UK suspension (Information Commissioner's Office, 2024).

LinkedIn's parent company Microsoft had strong incentives to use LinkedIn's professional data corpus: it is among the highest-signal professional knowledge datasets in existence, reflecting real career trajectories, professional writing, industry discussions, and organizational structures at a fidelity that no public dataset approaches.

3.4 YouTube / Google

YouTube's October 2023 Terms of Service update added explicit language permitting Google to use content uploaded to YouTube "to improve and develop Google products and services, including machine learning technologies and AI." The update did not offer a content-creator opt-out; existing content already uploaded was covered.

For Google, YouTube data represents an asset of unusual multimodal richness: video, audio, speech, captions, transcripts, user comments, engagement metrics, and channel metadata. DeepMind's research has used YouTube video data for training audio and video models (Conformer speech recognition, video language models). Google's Gemini multimodal model series builds on training infrastructure that includes YouTube-derived video-text pairs, though the exact composition has not been disclosed.

The consent architecture is notable: YouTube's Partner Program distributes a share of advertising revenue to creators, but this revenue share is based on views, not on the value of the creator's content as training data. A creator whose technical tutorials are extensively used to train a coding AI model receives no additional compensation beyond whatever advertising revenue their views generate.

3.5 TikTok / ByteDance

TikTok occupies a structurally unusual position: it operates under Chinese ownership (ByteDance) while serving predominantly Western users, creating a regulatory asymmetry where the data practices applicable to the majority of users (US, EU, Southeast Asia) are governed by the parent company's data infrastructure and strategic priorities.

ByteDance has disclosed that it uses TikTok user data, including video content, behavioral signals, and interaction data, to train its recommendation models. Less publicly documented is the extent to which this data informs ByteDance's broader AI research, which includes Doubao (ByteDance's LLM), video generation models, and speech synthesis systems that are deployed in the Chinese market.

The Irish DPC imposed a €345 million fine on TikTok in September 2023 for violations of children's data protection under GDPR — the first major enforcement action against the platform in the EU. However, the fine addressed consent mechanisms for minors, not the broader AI training data use question.

The strategic significance of TikTok's data for AI training extends beyond text: its video corpus, especially short-form video with associated audio, engagement patterns, and creator demographics, is among the most comprehensive multimodal behavioral datasets in existence.

3.6 Reddit

Reddit presents a contrasting governance model to the unilateral ToS amendments of other platforms: it has moved toward an **explicit licensing market** for its data corpus.

February 2024: Reuters reported (and Reddit publicly acknowledged in its S-1 prospectus) that Reddit had signed a data licensing deal with Google valued at approximately \$60 million per year, granting Google access to Reddit's content for AI training purposes shortly before Reddit's March 2024 IPO (Reuters, 2024; Reddit, Inc., 2024). Reddit's S-1 disclosed an aggregate \$203 million in AI data-licensing revenue under contracts with terms of two to three years. This deal made Reddit one of the first social platforms to commercially acknowledge and price the AI training value of its user-generated corpus.

Reddit's Data API pricing change in June 2023 — which priced out the third-party apps that many users relied on, precipitating a major platform protest — was explicitly motivated in part by the need to capture value from AI companies scraping Reddit data without payment. Reddit CEO Steve Huffman framed the issue in a 2023 *New York Times* interview: "The Reddit corpus of data is really valuable... But we don't need to give all of that value to some of the largest companies in the world for free" (Isaac, 2023).

The Reddit model — licensing rather than unilateral extraction — represents a different point on the consent spectrum, though it still does not distribute proceeds to the individual users whose contributions constitute the asset being licensed.

§4 The Consent Architecture and Its Failures

The legal mechanism through which social platforms claim user consent for AI training is the **Terms of Service agreement** — a contract of adhesion that users must accept to use the platform, presented as a take-it-or-leave-it condition with no meaningful negotiation and, typically, retroactive application to previously contributed content.

Under EU GDPR, consent must be freely given, specific, informed, and unambiguous (Article 7). The Article 29 Working Party (now EDPB) has long maintained that ToS-bundled consent does not meet this standard for secondary uses not essential to the primary service. AI training is not essential to the provision of a social

network; it is a secondary commercial use that could — in principle — be governed by a separate consent mechanism.

The Irish DPC's enforcement pause on Meta's EU AI training rested on precisely this analysis: Meta's "legitimate interests" legal basis was contested, and the scale and nature of the secondary use was held to require specific assessment. The DPC's intervention established a precedent — since applied by multiple other EU supervisory authorities — that **foundation model training on platform data is not automatically covered by existing social platform consent structures**.

The opt-out default, illustrated by LinkedIn's August 2023 episode, represents a weaker consent mechanism: users are nominally given a choice, but the choice is buried in settings, enabled by default, and applied to data already contributed. Under GDPR's consent requirements, opt-out defaults for secondary purposes are generally held insufficient; the data protection community has converged on the position that AI training requires opt-in consent or a clearly established legitimate interests basis that survives the balancing test.

In the United States, no equivalent consent framework exists at federal level. The FTC's "Trade Regulation Rule on Commercial Surveillance and Data Security" advance notice of proposed rulemaking (Federal Trade Commission, 2022 — 87 Fed. Reg. 51273, 22 August 2022) identified behavioral data collection and secondary use as concerns and explicitly raised whether the FTC should regulate the use of personal data for algorithmic decision-making and AI training, but rulemaking has not progressed to binding rules.

§5 The Second Extraction

This chapter's central argument connects to the extraction analysis of Draft F: the social platform data situation constitutes a **second layer of extraction** that is structurally analogous to but distinct from the open-commons extraction.

The open-commons extraction (Draft F, Draft C) involves:

- Data that contributors *deliberately made public* for knowledge-sharing
- Platforms (Common Crawl, GitHub, Wikipedia) with no commercial AI interest
- Extraction by third parties (AI labs) outside the platform relationship

The social platform extraction involves:

- Data that users shared *within a platform relationship*, with varying privacy expectations
- Platforms that *are themselves* building or integrating AI systems
- Extraction by the same entity that provides the service — creating a fundamentally different conflict of interest

Zuboff's (2019) concept of **behavioral surplus** — the data collected beyond what is needed to provide the service, repurposed as raw material for prediction products — captures the structural dynamic. Social platforms collect behavioral surplus as a matter of business design; AI training is the most recent and most capital-intensive application of that surplus. The value asymmetry is acute: users generate the behavioral surplus through their social activity; the platform captures the surplus and converts it to AI capability; users bear the risks of systems trained on their data (discriminatory targeting, manipulation, behavioral prediction) while receiving no share of the value created.

Zuboff's framework was developed primarily around behavioral targeting for advertising. The transition to AI foundation model training extends the surplus extraction in two ways: (1) it transfers value outside the platform relationship to models deployed in entirely different contexts; (2) it involves a qualitative change in the capability of the derived product — not merely predicting which advertisement a user will click, but training a general-purpose system that can reason, generate, and act across domains.

§6 Governance Responses and Design Proposals

Transparency requirements. The EU AI Act's GPAI provisions require disclosure of training data used to train general-purpose AI models above a compute threshold, including whether data was obtained from rights holders with permission (GPAI Code of Practice, 2024). Applied to social platform foundation models, this requirement would necessitate disclosure of the categories and volumes of user-generated content used in training — disclosure that no major platform has voluntarily provided at this level of specificity.

Opt-in rather than opt-out. The data protection consensus — reflected in the EDPB's guidance on AI training under GDPR (2024) and in DPC enforcement actions — is that AI training on personal data requires opt-in consent or a rigorously applied legitimate interests basis that accounts for the reasonable expectations of the data subject. Platform designs that make opt-in the default would substantially reduce the available training data pool, creating genuine tension between regulatory compliance and foundation model capability.

Data cooperatives and revenue sharing. The Reddit licensing model — whatever its limitations — demonstrates that the AI training value of social platform data can be priced and distributed. A more equitable variant would distribute licensing proceeds to individual contributors proportionate to their contribution to the licensed corpus. Proposals for data cooperatives (Sandoval, 2020; Birch, Cochrane & Ward, 2021) would give users collective bargaining power over the terms on which their data is used for AI training, analogous to the collective licensing frameworks that govern music and photography.

Temporal and contextual consent. User consent at account creation cannot meaningfully anticipate the AI capabilities that will exist ten years later; content contributed in 2010 could not have been contributed in the context of 2025 foundation model training. A principle of **temporal integrity** — that consent must be renewed for new secondary uses that were not foreseeable at the time of original contribution — would require affirmative user notification and choice before historical data is used for AI training.

Federated training with differential privacy. Technical approaches to training on user data without centralizing the data itself — federated learning with differential privacy guarantees — are mature enough to be deployed at scale (McMahan et al., 2017). Google's Gboard keyboard uses federated learning for next-word prediction; the same architecture could in principle be applied to social platform foundation models. The tradeoff is reduced model capability; current frontier models appear to require centralized data at scale that federated approaches cannot replicate.

§7 Open Questions

1. **What is the legal status of model weights trained on GDPR-regulated data?** If Meta's EU user data was used for Llama training without adequate legal basis, what is the remedy — deletion of the trained model, prohibition on its deployment in the EU, financial penalty? The GDPR's "right to erasure" and the technical impossibility of "machine unlearning" at model scale create a genuine legal gap (Zhang et al., 2023/2024; cf. Draft C §6).
2. **Does the Reddit licensing model scale?** If social platforms transition to licensing their user-generated corpora to AI companies, what governance structures ensure that proceeds are distributed to contributors rather than captured entirely by the platform? What is the legal relationship between the platform as licensor and the user as original creator of the licensed content?
3. **How does TikTok's multimodal corpus compare in AI training value to text-dominated platforms?** Short-form video with behavioral engagement signals may constitute the highest-signal multimodal training dataset in existence for human behavior modeling. The strategic AI implications of ByteDance's access to this corpus have not been systematically analyzed.

4. **What is the relationship between platform data and model alignment?** RLHF — the primary alignment technique for current LLMs — is trained on human preference data. The most scalable source of human preference signals at fine granularity is social platform behavioral data (likes, shares, engagement duration, explicit reactions). Are the largest social platforms effectively the unacknowledged alignment infrastructure for frontier AI?
5. **Can users recover agency after behavioral surplus extraction?** Once a model is trained on a user's behavioral history, deleting the user's account does not delete the learned representations. Conceptually, the user's behavioral signature persists in the model indefinitely. What rights, if any, do users have over models trained on their data — and are these rights technologically meaningful or merely nominal?
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Changelog

See `Chapter drafts/changelogs/av-changelog.md` for full iteration history.